We consider a smooth two–dimensional function

$$f(x,y) = \frac{1}{1 + \left\| C \begin{pmatrix} x \\ y \end{pmatrix} \right\|^2},$$

where $C \in \mathbb{R}^{2\times 2}$ controls anisotropy.

The function admits a Chebyshev expansion

$$f(x,y) = \sum_{k=0}^{d_x} \sum_{\ell=0}^{d_y} C^{\text{true}}_{k\ell}\, T_k(x)\, T_\ell(y),$$

where T_k denotes the Chebyshev polynomial of degree k and C^{true} is the reference coefficient matrix obtained by projection onto the tensor-product basis.

For each configuration $s = 1, \dots, S$, we generate sets of points

$$\{x_i^{(s)}\}_{i=1}^{n_x}, \qquad \{y_j^{(s)}\}_{j=1}^{n_y},$$

and define the scalar energy as a mean over pair interactions

$$E^{(s)} = \frac{1}{n_x n_y} \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} f\big(x_i^{(s)}, y_j^{(s)}\big).$$

Define pooled ACE features

$$a_x^{(s)}(k) = \sum_{i=1}^{n_x} T_k\big(x_i^{(s)}\big), \qquad a_y^{(s)}(\ell) = \sum_{j=1}^{n_y} T_\ell\big(y_j^{(s)}\big).$$

By linearity of the expansion, the energy can be written as

$$E^{(s)} = \left(a_x^{(s)}\right)^\top \left(\frac{1}{n_x n_y} C^{\text{true}}\right) a_y^{(s)}.$$

We therefore fit a bilinear model

$$E^{(s)} \approx \left(a_x^{(s)}\right)^\top C a_y^{(s)},$$

where the coefficient matrix is parameterized in Tucker form

$$C \approx A G B^\top,$$

with

$$A \in \mathbb{R}^{(d_x+1)\times R_x}, \quad B \in \mathbb{R}^{(d_y+1)\times R_y}, \quad G \in \mathbb{R}^{R_x \times R_y}.$$

Introducing reduced features

$$u^{(s)} = (a_x^{(s)})^\top A, \qquad v^{(s)} = (a_y^{(s)})^\top B,$$

the model becomes

$$E^{(s)} \approx (u^{(s)})^\top G\, v^{(s)}.$$

The parameters A, B, and G are computed by alternating least squares (ALS). At each iteration, we:

- Fix B, solve a linear least squares problem for an intermediate matrix $\widetilde{G}$, and compute its truncated SVD to update A and G.
- Fix A, solve an analogous least squares problem for $\widetilde{H}$, and use its truncated SVD to update B and G.

After convergence, the learned tensor is

$$C_{\text{learned}} = AGB^{\top}.$$

Since the energy was generated from $f(x, y)$, we compare against the reference tensor

$$C_{\text{ref}} = \frac{1}{n_x n_y} C^{\text{true}},$$

and report the coefficient error

$$\|C_{\text{ref}} - C_{\text{learned}}\|_2.$$

In summary, the code verifies that ACE-style pooled features combined with a low-rank Tucker parameterization recover the underlying Chebyshev coefficient tensor of the interaction function.